

Full length article

Pandanus tectorius fruit extract promotes Hsp70 accumulation, immune-related genes expression and *Vibrio parahaemolyticus* tolerance in the white-leg shrimp *Penaeus vannamei*

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ABSTRACT

Plants and herbal extracts are indispensable for controlling the spread of disease-causing bacteria, including those that infect aquatic organisms used in aquaculture. The use of plant or herbal extract is expected to be safe for aquatic animals and less harmful to the environment, as opposed to conventional therapeutic alternatives such as antibiotics that promote the occurrence of potential antibiotic-resistant bacteria when used improperly. The efficacy of *Pandanus tectorius* fruit extract in the regulation of Hsp70 expression, pro-phenoloxidase (ProPO), peroxinectin, penaeidin, crustin and transglutaminase, all immune peptides essential for *Vibrio* tolerance in white leg shrimp, *Penaeus vannamei*, was investigated in this study, which included the determination of the safety levels of the extract. Tolerance of shrimp against *Vibrio parahaemolyticus*, a pathogenic bacteria that causes Acute Hepatopancreas Necrosis Disease (AHPND), was assessed on the basis of median lethal dose challenge survival (LD50 = 10⁶ cells/ml). Mortality was not observed 24 h after exposure of 0.5–6 g/L of the fruit extract, indicating that *P. tectorius* was not toxic to shrimp at these concentrations. A 24-h incubation of 2–6 g/L of the fruit extract increased shrimp tolerance to *V. parahaemolyticus*, with survival doubled when the maximum dose tested in this study was used. Concomitant with a rise in survival was the increase in immune-related proteins, with Hsp70, ProPO, peroxinectin, penaeidin, crustin and transglutaminase increased 10, 11, 11, 0.4, 8 and 13-fold respectively. Histological examination of the hepatopancreas and muscle tissues of *Vibrio*-infected shrimp primed with *P. tectorius* extract revealed reduced signs of histopathological degeneration, possibly due to the accumulation of Hsp70, a molecular chaperone crucial to cellular protein folding, tissue repair and immune response of living organisms, including Penaeid shrimp.

1. Introduction

Penaeid shrimps are among the most traded seafood products in the world, with a value of approximately US\$32 billion in 2017 [1]. Specifically, the demand for *P. vannamei*, commonly referred to as White-leg shrimp, is particularly favourable as it grows faster compared to other shrimp species. Unlike *Penaeus monodon* [2,3] and *Penaeus stylirostris* [4], *P. vannamei* is capable of tolerating a wide range of salinity, making

it easier to maintain even inland areas [5]. Alongside the rapid increase in the production of *P. vannamei*, there are several problems [6,7] with disease which constitute a major obstacle to the development of the industry [8,9]. Vibriosis is the most serious of all known diseases, causing massive deaths of Penaeid shrimps worldwide [10–13].

Vibriosis, a disease caused by *Vibrio* bacteria, affects many species of fish and shellfish, including *P. vannamei* [14]. Some *Vibrio* species are pathogenic, while others remain opportunistic, threatening only when

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host immunity is compromised by biotic or abiotic stressors. Infection occurs when shrimps are weak, with symptoms varying depending on the level of infection and the type of *Vibrio* that infects them. As an example, shrimp infected with *V. parahaemolyticus* appears hypoxic at an early stage, and when the infection is severe, the diseased shrimp stops feeding and reddening of the body tissues occur, with the gills eventually turning red or brown [15]. Histological examination may reveal the formation of septic haemocytic nodules in the muscle and hepatopancreas [16], typical signs of inflammation. Any damage to the hepatopancreas shrimp often results in mortality, mainly because the hepatopancreas organ is essential for the detoxification process and is involved in protein and hormone secretion [17,18]. Vibriosis has caused massive financial losses to farmers over the last decades. For example, the AHPND outbreak, which was first reported in Southern China in 2009, causes global economic losses in the shrimp farming industry of more than US\$1 billion per year [7]. Several methods have been suggested for controlling vibriosis in aquaculture, with the use of antibiotics perhaps the most effective, but with dire consequences for human health and the environment [13,20,21]. In this context, the continuous use of antibiotics and chemicals has led to the spread of antibiotic-resistant pathogens and has resulted in the bioaccumulation of chemicals in shrimp tissues [22]. These disadvantages have stimulated global efforts to discover biosafe alternatives to control vibriosis in aquaculture [23].

Plants are cheap and reliable natural resources for disease control in aquatic organisms, many of which are rich in phytochemicals capable of attenuating pathogenic microbes [24]. Others may have bioactive compounds that activate the host immune system [25,26], reduce inflammation [27], remove toxins [28,29], prevent tumour or cancer growth [30] and stress [31,32]. For example, the use of Indian Almond *Terminalia catappa* leave extract was effective in reducing bacterial disease in Siamese fighting fish *Betta splendens* [33] and promotes survival during gourami larviculture [34], possibly due to the high content of tannins, flavonoids and other potent bioactive anti-bacterial and anti-oxidant compounds. Tex-OE®, a prickly pear skin extract, *Opuntia ficus indica* has also been shown to consist phytochemicals capable of inducing Hsp70 synthesis in different test animals under non-stressful conditions, thus improving their survival during disease [35–38].

The efficacy of *P. tectorius* fruit extracts on *P. vannamei* tolerance during *V. parahaemolyticus* challenge was examined in this study. *P. tectorius*, or commonly referred to as thatch screwpine, is a coastal mangrove plant native to Malaysia, Eastern Australia and the Pacific Islands. *P. tectorius* belongs to the Pandanaceae family and is closely related to *Pandanus odoratissimus*, *Pandanus fascicularis* and *Pandanus amaryllifolius* [39]. The fruits of *P. tectorius* are edible and resemble the shape of pineapples. Because the fruit and leaves are rich in anti-oxidant, anti-bacterial and anti-diabetic properties, they are traditionally used as a medicinal alternative for the treatment of infection, inflammation and other human diseases [40]. Specifically, the safety level of *P. tectorius* fruit extract was first determined, followed by a verification of the median lethal concentration (LD50) of *V. parahaemolyticus* on the basis of a standardised challenge test as described by Loc et al. (41). Upon verification of *Vibrio* LD50, the efficacy of *P. tectorius* extract in protecting *P. vannamei* against *V. parahaemolyticus* was determined. The effects of *P. tectorius* fruit extract on several key *P. vannamei* immune-related gene expressions, namely Hsp70, crustin, ProPO, penaeidin, peroxinectin and transglutaminase [42–53], were determined. The histopathology of post-larvae shrimp muscle and hepatopancreas exposed to *P. tectorius* fruit extract prior to the *Vibrio* challenge was determined. The degree of severity of *V. parahaemolyticus* to shrimp cell damage was verified and compared to non-exposed controls. The protective effects of *P. tectorius* fruit extract were discussed on the basis of the *Vibrio* tolerance capability.

2. Materials and methods

2.1. Maintenance of *P. vannamei*

P. vannamei post-larvae was purchased and acclimatised at 28 °C, 38 g/L salinity and 7.8–8.0 pH for two weeks prior to use in the experiment. Post larvae were kept at 2000 individuals/L and fed to satiation twice daily with *Artemia* nauplii. The quality of water was maintained by changing the rearing water once every two days. The health status of shrimps was monitored daily and only healthy animals measuring approximately 1 cm were selected and used for experiments.

2.2. Establishment of *V. parahaemolyticus* challenge test

The strain of *V. parahaemolyticus* used in this study was isolated from diseased shrimps in Vietnam [41]. *V. parahaemolyticus* from stock cultures stored at –80 °C were grown in marine agar and single colonies were transferred to marine broth for upscale. Bacteria were grown overnight in a shaker incubator, with temperatures maintained at 28 °C. Cells were harvested by centrifugation at 11200×g for 15 min and the pellets washed 3 times, each with 30 ml of filtered autoclaved seawater (FASW). *Vibrio* density was determined according to the MacFarland standard as essentially described [41,54] and challenge test was performed by incubation of 30 shrimps in separate 1 L aquaria containing 1×10^1 , 1×10^2 , 1×10^3 , 1×10^4 , 1×10^5 , 1×10^6 , and 1×10^7 of *V. parahaemolyticus*/ml (cells/ml). The lethal dose (LD50) resulting in 50% shrimp mortality was determined and the survival percentage was calculated [41]. Shrimps were not fed during the *Vibrio* challenge and the animals maintained in the absence of *Vibrio* were used as control. Challenge tests were repeated once, with each treatment being performed in triplicates.

2.3. Preparation of *P. tectorius* fruit extract and determination of the safety level

Fruit from *P. tectorius* was collected from Setiu Wetlands, Terengganu, Malaysia and transferred to the Natural Product Laboratory of the Institute of Marine Biotechnology, UMT. Fruit was peeled, cleaned and cut into small pieces. Fruit cuttings were air-dried and stored at 80 °C for the day before lyophilization (EYELA FD-550, USA). Dry samples were ground to fine powder using an electronic blender. Thirty grams of the powder were mixed with 300 ml of methanol in a round bottom flask and incubated at room temperature (RT) for 24 h. The extract was filtered through a Whatman No. 1 filter paper and subsequently concentrated in a vacuum using a rotary evaporator (BUCHI R-300, Switzerland), with a pressure and temperature of 337 mbar and 40 °C, respectively. The extracts with methanol were stored at RT prior to use in the experiments.

The safety level assessment of *P. tectorius* was performed by separate exposure of 30 shrimps to 0.5, 1, 2, 3, 4, 5 and 6 g/L of fruit extract, with non-exposed animals being used as control. Mortality was determined by counting dead shrimps 24 h after exposure. The assays were repeated once, with each treatment in triplicates.

2.4. Determination of *P. vannamei* tolerance against *V. parahaemolyticus* upon incubation in *P. tectorius* fruit extract

Thirty *P. vannamei* post-larvae exposed to a 24 h incubation of 0.5, 1, 2, 3, 4, 5 and 6 g/L of *P. tectorius* fruit extract were treated with 1×10^6 cells/ml, the median lethal dose (LD50) of *V. parahaemolyticus*. Shrimp unexposed to *P. tectorius* fruit extract was used as a control. Shrimp survival was determined 24 h after the *Vibrio* challenge by actively counting swimming animals. Moribund shrimps were regarded as dead. The experiments were repeated once, with each treatment being performed in triplicates.

2.5. Hepatopancreas and muscle tissue histology profiles

Hepatopancreatic and control shrimp muscle tissues (without incubation of *P. tectorius* fruit extract and *Vibrio* exposure) and *V. parahaemolyticus*-challenged shrimps (with or without incubation of *P. tectorius* fruit extract) were fixed in the Davison reagent for 48 h and subsequently embedded in paraffin (melting point 54–56 °C). Embedded tissues were placed on a cold plate (LeicaTM EG1150C, Germany) and cut with an automated microtome (LeicaTM RM2255 Microtome, Germany) at a thickness of 5 µm [55]. Samples were then stained with Harris hematoxylin and eosin (H&E). Slides were fixed by dibutylphthalate polystyrene xylene (DPX) and histological images were analysed using the LeicaTM DM LB2 light microscope (40 × magnification) and the LeicaTM Image Analyzer System (Germany) respectively.

2.6. Determination of Hsp70 expression in shrimp

One hundred milligrams of the caudal muscle tissues from shrimps exposed to different concentrations of *P. tectorius* fruit extract were collected and rinsed several times with distilled water. Protein extraction was performed as described by Sung et al. [56] with minor modification. Briefly, tissues were homogenized in cold Buffer K (150 mM sorbitol, 70 mM potassium gluconate, 5 mM MgCl₂, 5 mM NaH₂PO₄, 40 mM HEPES, pH 7.4) [54,57] and Protease Inhibitor Cocktail (Sigma-Aldrich Inc-P8849, USA) as recommended by the manufacturer. The homogenates were centrifuged at 4 °C for 30 s and the supernatant containing the crude protein was collected in a sterile tube. The concentration of protein was determined using the Bradford method prior to electrophoresis [58].

SDS polyacrylamide gel electrophoresis was performed as described by Loc et al. [41]. Five microliters of the homogenates were combined with equal volume of sample buffer, vortexed and heated at 95 °C for 5 min. The samples were cooled and centrifuged at 2200×g for 1 min. Ten microliter samples representing 1.0 mg wet shrimp weight were loaded with 10% polyacrylamide gels in each lane, and electrophoresis was resolved at 120 V for 15 min, followed by 150 V for 45 min. Two gels were run simultaneously with one stained with Coomassie Biosafe (BioRad Laboratories, USA) and the other blotted to polyvinylidene fluoride transfer membrane (BioRadImmun-Blot™ PVDF, USA) for Western immunoblotting.

Membranes were incubated in blocking buffer (50 ml of phosphate-buffer saline containing 0.2% (v/v) Tween-20 and 5% (w/v) bovine serum albumin) at 25 °C for 60 min and subsequently incubated with mouse monoclonal antibody to Hsp70 (Clone 3A3, Thermo Scientific, USA) diluted 1:5000 in PBS. Goat anti-Mouse IgG F(ab')₂ polyclonal antibody conjugated with horse radish peroxidase (HRP) (Bioreagent-SAB-100 J, Stressgen, Canada) was used as secondary antibody at a dilution of 1:2000 in PBS. For protein detection, 0.7 mM diaminobenzidine tetrahydrochloride dehydrate (DAB) was used as substrate in combination with 0.01% (v/v) H₂O₂ in 0.1 M Tris-HCL (pH 7.6) [56, 59].

2.7. Densitometry analysis

The Hsp70 expression was determined using a densitometry analysis. Blots were scanned using a densitometer (GS-800 BioRad Laboratories, USA) and quantification was performed by measuring the Hsp70 band intensity with Quantity One software (BioRad Laboratories, USA). The expressed protein was interpreted as reflective density/mm² and presented as a fold difference.

2.8. Quantification of Immune-Related Genes by qRT-PCR

One hundred milligrams of muscle tissues from control shrimps and *V. parahaemolyticus*-challenged shrimps (with or without *P. tectorius* fruit extract incubation) were used for RNA extraction. RNA was extracted

according to the TRIsure™ kit manufacturer's protocol (Bioline, UK) and the purity and quantity were spectrophotometrically determined at 260 and 280 nm. The first strand of cDNA was obtained from 1 mg of RNA by 45 min incubation using the cDNA synthesis kit (Bioline, UK). PCR was performed in the absence of a reverse transcript to confirm the lack of DNA transfer.

qRT-PCR was performed in the CFX Connect™ System (Bio-Rad, USA) using 2 × SensiMix SYBR No-ROX kit (Bioline, UK) with forward and reverse primers specific to Pro-PO, peroxinectin, penaeidin, crustin, haemocyanin, Hsp70 and β-Actin, the latter being used as a house-keeping gene (Table 1). The amplification was performed at 95 °C for 10 min followed by 40 cycles at 95 °C for 15 s and 58 °C for 1 min. For Hsp70, however, 40 cycles of 95 °C for 5s and 60 °C for 31s have been adopted [41]. The melting curve analysis of the products of the PCR was performed to confirm the amplification of a single product. Cycle threshold (CT) values were determined using CFX Manager software (Bio-Rad, USA). The fold difference for each of the immune-related genes relative to β-actin was calculated using the 2^{-ΔΔC_t} method described by Clegg et al. [57] and Sinnasamy et al. [60].

2.9. Statistical analysis

Descriptive statistics of survival were analysed using Minitab 14® computer software. Data were tested for normality and homogeneity of variance prior to Analysis of Variance (ANOVA). Data with significant differences (p ≤ 0.05) were separated using Fisher's least significant difference (LSD) and Kruskal-Wallis non-parametric test was performed when the assumption of normality and homogeneity of the data were insignificant. The statistical differences in gene expression were also analysed using a similar method. The results were shown in bar charts using GraphPad Prism (version 5.01) software.

3. Results

3.1. Pre-treatment with methanolic *P. tectorius* fruit extract increased the survival of *P. vannamei* against *V. parahaemolyticus*

The survival percentages of *P. vannamei* exposed to varying doses of *V. parahaemolyticus* and *P. tectorius* fruit extract are shown in Figs. 1 and 2, respectively. Shrimp mortality increased concomitantly with the increase in *Vibrio* concentration during challenge, with LC₅₀ occurring at 1 × 10⁶ cells/ml (Fig. 1). Survival was approximately 96% when shrimp post larvae were exposed 24 h to different concentrations of *P. tectorius* fruit extract, similar to control treatment. The value of LC50 was estimated at 1.58 × 10⁶ g/L (Fig. 2). Shrimp incubated with *P. tectorius* fruit extract showed higher survival at 48 h LC50 *V. parahaemolyticus* challenge, with the number of viable animals increased from 40 to 80% when the concentrations of fruit extract used to incubate shrimp post

Table 1

Primers specific for Hsp70, Crustin, Prophenoloxidase, Penaeidin, Peroxinectin, Transglutaminase and β-actin.

Primers	Accession #	Sequence
Hsp70 (F)	EF495128	5'-CCTCCAGGACTCTCTCAACG-3'
Hsp70 (R)		5'-GGTCACGTCCAACAGCAAC-3'
Crustin (F)	AF430076	5'-ACGAGGCAACCATGAAGG-3'
Crustin (R)		5'-AACCACCACCAACACCTAC-3'
Prophenoloxidase (F)	AY723296	5'-CGGTGACAAAGTTCTCTTC-3'
Prophenoloxidase (R)		5'-GCAGGTGCGCGTAGTAAG-3'
Penaeidin-3a (F)	Y14926	5'-CACCCCTTCGTGAGACCTTTG-3'
Penaeidin-3a (R)		5'-AATATCCCTTTCCACGTGAC-3'
Peroxinectin (F)	AF188840	5'-CGAAGCTTCTTGCAACTACCA-3'
Peroxinectin (R)		5'-GCAGGCTGATTAAACTGGCTT-3'
Transglutaminase (F)	EF081004	5'-GAGCTTCAAGATCGAGGATCGA-3'
Transglutaminase (R)		5'-GCTGGTGTTCGTAGCGGTTATC-3'
β-actin (F)	AF300705	5'-CCACGAGACCACCTACAAC-3'
β-actin (R)		5'-AGCGAGGGCAGTGATTTC-3'

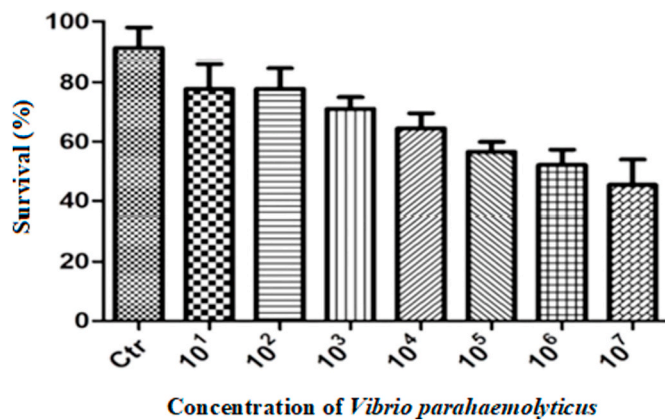


Fig. 1. Survival of *P. vannamei* after 24 h exposure to *V. parahaemolyticus*. Ctr, control (*P. vannamei* not exposed to *V. parahaemolyticus*); Treatments, 1×10^1 to 1×10^7 cells/ml. Error bars are standard error. $LC_{50} = 1 \times 10^6$ cells/ml ($r = 0.991$; $p \leq 0.05$).

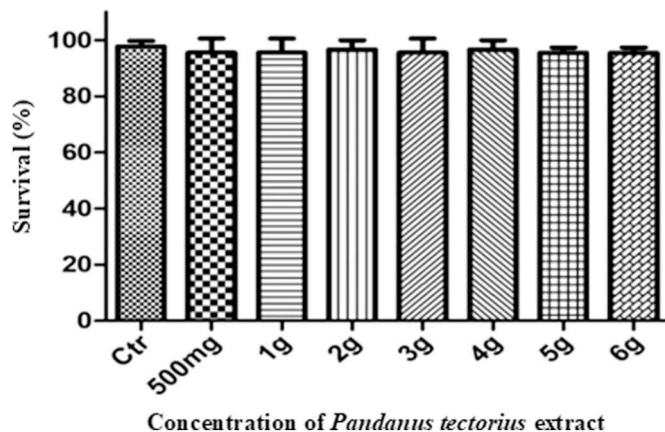


Fig. 2. Survival of *P. vannamei* after 24 h exposure to different concentrations of *P. tectorius* fruit extract. Ctr, control (*P. vannamei* not exposed to fruit extract); Treatments, 0.5–6 g/L. Error bars are standard error. $LC_{50} = 1.58 \times 10^6$ g/L ($r = 0.991$; $p \leq 0.05$).

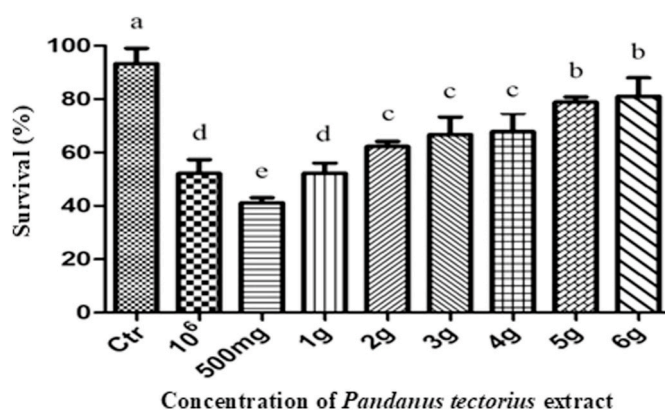


Fig. 3. Survival of *P. vannamei* after pre-treatment with *P. tectorius* fruit extract and 24 h exposure to 1×10^6 cells/ml *V. parahaemolyticus*. Ctr, control (*P. vannamei* not exposed to fruit extract); shrimp treated with *V. parahaemolyticus* alone (10^6); Pre-treatments, 0.5–6 g/L. Error bars are standard error. Bars with different lower case letters differ significantly ($P < 0.05$).

larvae increased from 0.5 g to 6 g (Fig. 3).

P. tectorius Exposure Enhanced the Expression of Hsp70 and Immune-Related Genes in *P. vannamei*.

As revealed by SDS-PAGE, the intensity of the *L. vannamei* 70 kDa protein band increased upon exposure to methanolic *P. tectorius* fruit extract (Fig. 4A). Western blot immunoblotting with anti-Hsp70 antibody revealed a significant increase in the expression of shrimp Hsp70 (Fig. 4B), with densitometry analysis indicating an increase from 1.3 to 20.6-fold when the concentrations of extract used in incubation increased from 0.5 g to 6 g (Fig. 5) (Table 2). On other aspect, quantification of the immune-related genes expression by qPCR revealed a 10, 11, 11, 0.4, 8 and 13-fold increase in Hsp70, ProPO, peroxinectin, penaeidin, crustin and transglutaminase mRNA, respectively upon exposure to the highest concentration of *P. tectorius* fruit extract (6 g/L) (Fig. 6).

Pre-treatment with methanolic *P. tectorius* Fruit Extract Protected Tissues of *P. vannamei* during *V. parahaemolyticus*-Challenged Shrimp.

Histological analysis of shrimp hepatopancreas exposed to fruit extract and then exposed to *V. parahaemolyticus* revealed intact tubules and tubule lumen similar to those of the positive control group (shrimps without *P. tectorius* or *V. parahaemolyticus* treatment) (Fig. 7). Muscle tissue fibre appeared to be normal without lesion or necrosis (Fig. 8). The negative control group (shrimps without *P. tectorius* treatment but challenged with *V. parahaemolyticus*) however displayed enlarged tubule lumen, hypertrophied nuclei with margined chromatin and sloughing of the hepatopancreas (Fig. 7). A coagulative necrosis in the skeletal muscle along with haemolytic infiltration and extensive loss of muscle structure were observed in this group. There was also an increase in vacuolization of the cells lining the tubule lumen epithelium in the positive control group.

4. Discussion

Vibriosis is a major disease problem in aquaculture, leading to severe mortality of shellfish and finfish worldwide, resulting in massive financial losses. *Vibrio*-related infections can occur in both hatcheries and ponds in Penaeid shrimp farming. Infection occurs when *Vibrio* enters the barrier and host. The bacteria replicate and attack the host after reaching sufficient numbers [19]. Common *Vibrio* species infecting

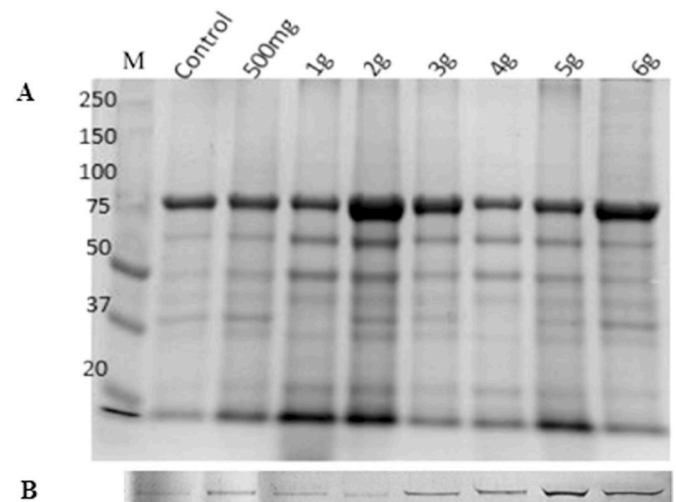


Fig. 4. Protein expression of *P. vannamei* by SDS-PAGE and Western blot after exposure to *P. tectorius* fruit extract for 24 h. (A) Lane 1, M (molecular weight marker); Lane 2, protein extracted from shrimps without any treatment (control); Lane 3–9, protein extracted from shrimps treated with different concentrations of fruit extract (0.5–6 g/L). Arrow indicates the estimated size of protein bands. (B) Western blot analysis of agarose gel. The expression of Hsp70 differ significantly between the control and treatments.

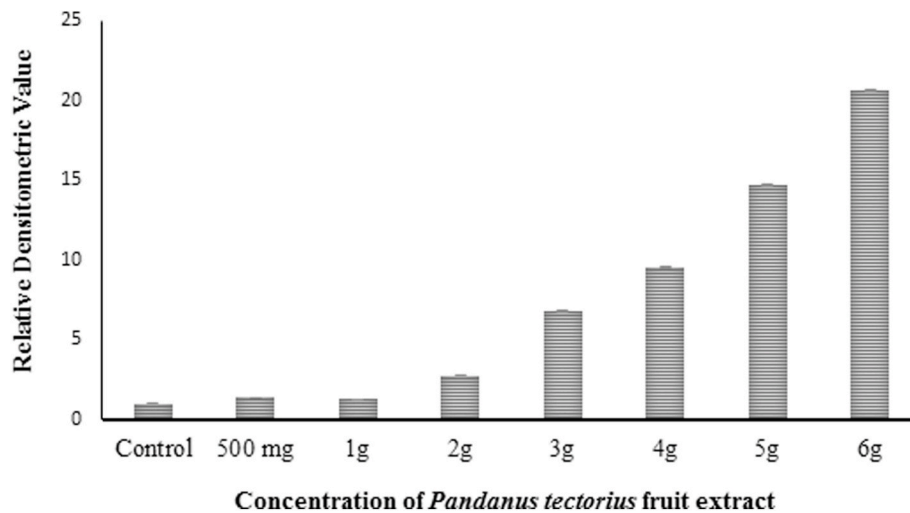


Fig. 5. Effect of *P. tectorius* fruit extract on Hsp70 expression of *P. vannamei*. Bar shows the relative density value developed by integrated densitometric values. Each bar represents the mean $\pm$ SEM.

Table 2

The densitometry data that shows the relative density (RD) of expression of Hsp70 in *P. vannamei* treated with methanolic fruit of *P. tectorius* for 24 h.

Treatment	Adj. Vol. (RD)	Relative Quantity
Control	13.60	1
500 mg	18.49	1.35
1g	17.00	1.25
2g	37.25	2.73
3g	91.81	6.74
4g	129.8	9.54
5g	199.21	14.64
6g	280.54	20.62

Penaeid shrimps include *V. harveyi*, *V. vulnificus*, *V. alginolyticus*, *V. penaeida* and *V. parahaemolyticus* [61,62], the latter which were used in this study. The median lethal dose of *V. parahaemolyticus* occurred in 1×10^6 cells/ml, with mortality apparent in all challenge tests performed. This observation strongly indicated that the strain of *V. parahaemolyticus* used in this study was pathogenic to *P. vannamei*. An earlier study had reported that 5×10^5 cells/ml *V. parahaemolyticus* was sufficient to kill 50% of *P. vannamei* juvenile within 96 h of challenge [63] whereas a higher concentration was required to produce similar results in the Tiger prawn *P. monodon* [64,65]. Total mortality of the experimental shrimp occurred at 1×10^8 cells/ml in 48 h. In this context, it should be noted that the virulence of *V. parahaemolyticus*

varies significantly from one strain to another and its ability to cause shrimp mortality depends primarily on the animal fitness and stage, the experimental conditions as well as the exposure time during the challenge tests. Over the years, the LD50 of *P. vannamei* with different species of *Vibrio* has been shown to occur for varying durations (i.e. 1–10 days) between 1×10^2 and 1×10^8 cells/ml [16,66,67].

The effects of *P. tectorius* fruit extract in the regulation of Hsp70 expression and other related immune genes for shrimp protection against *Vibrio* were investigated in this study. The use of plant materials as cheap and reliable natural resources for disease control in aquaculture is highly anticipated, but thorough testing is needed to determine whether they are safe for use. The toxicity of *P. tectorius* was determined here with results showing that the methanolic fruit extract is safe for *P. vannamei*. In this context, shrimp survival upon 24 h incubation remained high (above 95%) at the highest dose examined. Although there is little information on the use of *P. tectorius* in aquatic species, clinical trials with mice and rats revealed LD50 in excess of 2000 mg/kg [68–70], a concentration considered to be extremely high for any type of toxic agent tested on rodents. In addition, it was found that the levels of phenol, flavonoid, steroid, terpenoid, saponin and glycoside, all potential toxic phytochemical compounds in *P. tectorius* remained low and the prolonged exposure of methanol extract to mammalian cells did not cause any cell damage, indicating that they were not toxic [40]. However it has been reported that the 24 h LC₅₀ value of aqueous extract of *Pandanus odoratissimus*, a closely related species to *P. tectorius*, upon an in vitro lethality assay of the brine shrimp *Artemia* was approximately

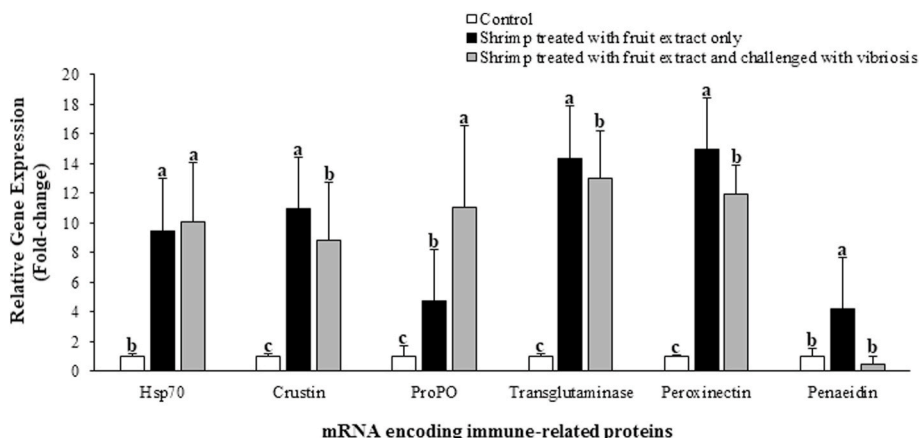


Fig. 6. Effect of *P. tectorius* fruit extract on immune genes expression of *P. vannamei*. The value at each point represents the fold change of mRNA expression (normalised against internal control β -actin) relative to the control value which are the samples from shrimps without any treatment. Statistical analysis was performed using Tukey's test. Bars with different lower case letters differ significantly ($P < 0.05$). The data shown are representative of two independent experiments with similar expression patterns.

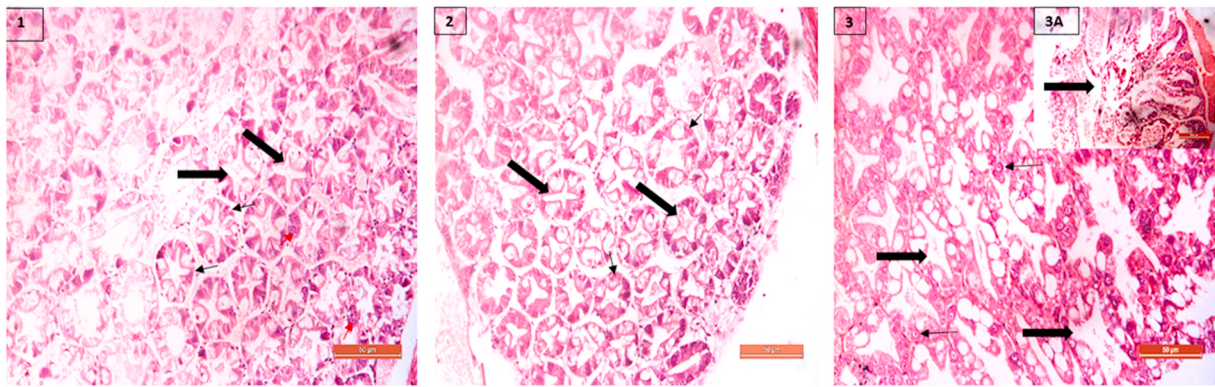


Fig. 7. Histological section of shrimp hepatopancreas. 1. Hepatopancreatic section of control shrimps showing healthy tubule (black thin arrow), tubule lumen (black wide arrow) and epithelial cells lining the tubules (Red thin arrow). 2. Hepatopancreatic session treated with fruit extract and then challenged with *V. parahaemolyticus* showing intact tubule (black thin arrow) and tubule lumen (black wide arrow). 3. Hepatopancreatic tubules from shrimp challenged with *V. parahaemolyticus* showing enlarged tubule lumen of hepatopancreas (black wide arrow), hypertrophied nuclei with margined chromatin (black thin arrow). 3A. Sloughing of hepatopancreas in infected shrimp (black wide arrow). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

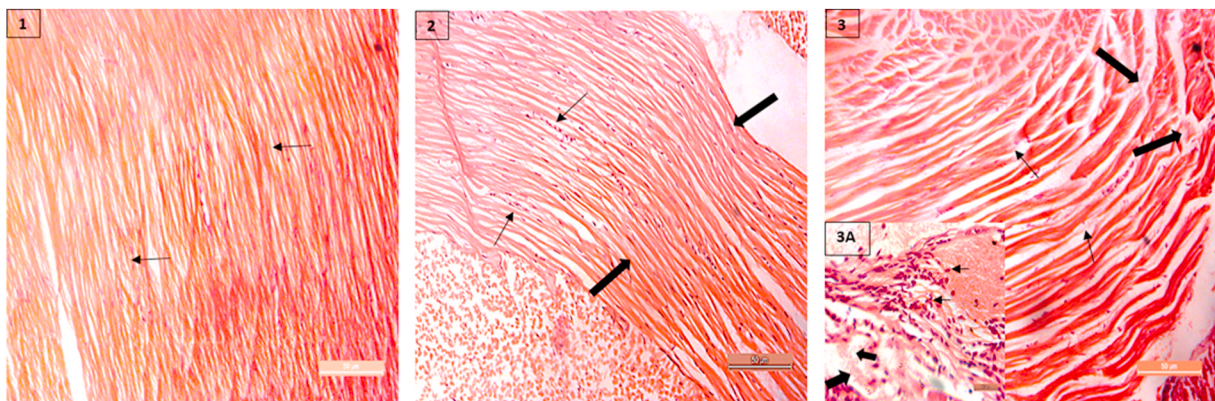


Fig. 8. Histological section of shrimp muscle. 1. Normal muscle from the control sample with muscle fibres intact (black thin arrow), no lesion and necrosis. 2. Muscle structure of shrimp treated with fruit extract and then challenged with *V. parahaemolyticus* showing no muscle necrosis (thick arrow) and slight hemocytic infiltration (black thin arrow). 3. Multifocal area of extensive coagulative necrosis (black wide arrow) and fragmentation of muscle fibres (black thin arrow) of *Vibrio*-challenged shrimp. 3A. A coagulative necrosis in skeletal muscle (wide arrow) along with hemocytic infiltration (thin arrow). Extensive loss of muscle structure.

33 µg/ml [71].

Interestingly, a 24 h incubation with methanolic *P. tectorius* fruit extract improved the survival of *P. vannamei* upon a median lethal dose of *V. parahaemolyticus*. Survival percentage reached 80%, a two-fold increase in the number of viable shrimp from the control treatment when exposure was performed at 6 g/L, the highest dose examined in this study. As a case in point, protection of *P. vannamei* against *V. parahaemolyticus* were enhanced in a dose dependant manner. This observation clearly indicated that methanolic *P. tectorius* fruit extract are beneficial to shrimp when added into the rearing water prior to pathogen exposure, with observations similar when extracts of *Sonneratia alba* (i.e. ethanoic, freshwater and saline water) [72], acetone extracts of *Syzygium aromaticum* and *Tamarindus indica*, and ethyl acetate extracts of *Emblia officinalis* [73] were applied on the larvae and juveniles of Tiger shrimp, *Penaeus monodon*.

At this point, it is difficult to speculate how methanolic *P. tectorius* fruit extract contributed to protection but like amla (*E. officinalis*), garlic (*Allium sativum*), clove (*S. aromaticum*) [73] and other herbs, the extracts consist compounds capable of stopping the activity or spread of pathogenic bacteria, including *Vibrio* [40]. Some compounds, on the other hand, activate the immune response of shrimp and induce the synthesis of potent immune proteins that destroy pathogens [31,74,75]. For example, α -phellandrene, a cyclic monoterpene phytochemical presence in herbal essential oils (such as those of *Schinus terebinthifolius*, *Anethum*

graveolens, *Moringa oleifera*, *Diplotaenia damavandica*, *Piper nigrum* etc.) has been shown to attenuate *V. alginolyticus* and application in the culture water enhanced the tolerance of *P. vannamei* towards pathogen challenge [31]. In a separate study, addition of the Tiwai onion, *Eleutheria americana* extract into the rearing water inhibited the growth of *V. harveyi* and prevented *P. monodon* mortality [76]. Additionally, extracts of lemon (*Citrus limon*), garlic (*A. sativum*), ginger (*Zingiber officinale*), Palm jaggery (*Borassus flabellifer*) and Black gram (*Vigna mungo*) were shown to be effective against microsporidian parasite, *Enterocytozoon hepatopenaei* (EHP), the causative agent of the white faeces syndrome in *P. vannamei*. Survival of shrimp was improved by 100% following exposure to these herbal extracts [77]. Likewise, a 48 h incubation with extract of the mangrove *S. alba* improved the survival of *P. monodon* post-larvae against *V. harveyi* and *Saprolegnia sp*, possibly due to immune activation [72].

Here, we have shown that the increased tolerance of *P. vannamei* to *V. parahaemolyticus* may be associated with the induction of Hsp70 caused by short-term incubation of *P. tectorius* methanolic fruit extract. In fact, the protective effects of Hsp70 on *P. vannamei* and other aquatic species during pathogen exposure have been demonstrated [54,59,60, 78,79]. In these studies, protection against the *Vibrio* challenge was shown to be correlated with the increase in Hsp70 expression, the latter induced by heat [80]. Heat shock treatments have been effective in promoting the induction and accumulation of Hsp70 shrimp, but this

method has been considered invasive, sometimes resulting in unprecedented mortality [35–38]. It is therefore crucial to identify non-invasive ways of substituting heat shock treatments, and the results of this study provided a novel and safe solution to enhance Hsp70 accretion without imposing significant stress on *P. vannamei*. As a case in point, the phenolic compound pyrogallol acted as an Hsp70 inducer in the brine shrimp *A. franciscana* and subsequently enhanced their tolerance against *V. harveyi*, a common pathogen for this crustacean species. Protection might be associated with activation of the prophenoloxidase and transglutaminase immune responses [81].

Apart from Hsp70, methanolic fruit extract of *P. tectorius* were demonstrated to induce the expression of crustin, prophenoloxidase, transglutaminase, proxinectin, and penaeidin, all immune-related genes crucial for pathogen removal in *P. vannamei*. Crustin functions in apoptosis and phagocytosis of shrimp hemocytes [82] and they consist a whey acid protein (WAP) domain which inhibit microbial growth [83, 84]. Reduction of this antimicrobial peptide has been linked to the increased mortality of *P. vannamei* upon *V. penaeicida* infection [85]. Prophenoloxidase play important roles in melanisation, cytotoxicity reaction, cell adhesion, encapsulation, haemocytes locomotion and phagocytosis whereas peroxinectin regulate exocytosis, promote encapsulation and mediate non-self-recognition during microbial infection [86–88]. Penaeidin, antimicrobial peptides with chitin-binding activity, occurs in *P. vannamei* granulocytes and they are released to eliminate microorganisms engulfed by granulocytes during phagocytosis [43,89]. Augmentation in the expression of these immune-related genes either by dietary immunostimulants or plant-based bioactive compounds have been associated with the increase in Penaeid shrimp tolerance against *Vibrio* [31,90,91].

Microbial infection aggravate muscle necrosis and haemolytic infiltration in the Penaeid shrimp [63,92–98]. As revealed by histology examination, the skeletal muscles of *V. parahaemolyticus*-compromised shrimp displayed these typical signs, along with enlarged tubule lumen, hypertrophied nuclei with margined chromatin and sloughing of the hepatopancreas. Cell vacuolization observed in the compromised shrimp has also been established as a morphological characteristic of dead tissues resulting from prolonged colonization by bacterial or viral infection as well as exposure to different natural and artificial low-molecular-weight compounds [99]. Here, we demonstrated that application of *P. tectorius* fruit extract prevented muscle necrosis in the infected shrimps, with condition comparable to the negative controls (non-challenged individuals). The mechanistic action(s) by which methanolic *P. tectorius* fruit extract alleviated histopathological degeneration in *V. parahaemolyticus* remained vague but Hsp70 accretion may promote wound healing in mammalian cells, providing rapid recovery of damaged tissues upon infection [100]. This calls for future work aimed at identifying bioactive compound(s) in *P. tectorius* fruit extracts responsible for regulating this protection mechanism and Hsp70 functions in *P. vannamei* during disease.

Regardless the outcomes, this study represents the first to reveal that a short term exposure to methanolic *P. tectorius* fruit extracts activated the immune response of *P. vannamei* and protected them against the deleterious effects of *V. parahaemolyticus*. This finding is potentially useful in assisting the formulation of strategies to protect *P. vannamei* and possibly other shrimp species from Vibriosis. The use of plant materials as cheap and reliable natural resources for disease control in aquaculture is highly anticipated and they are potential substitutes to antibiotics and chemical-based therapeutic alternatives used during aquaculture.

CRedit authorship contribution statement

Anupa Anirudhan: Conceptualization, Methodology, Formal analysis, Investigation, Validation, Project administration, Writing - original draft, Writing - review & editing. **Victor Tosin Okomoda:** Formal analysis, Writing - original draft, Writing - review & editing. **Mat Taib**

Mimi Iryani: Writing - original draft, Writing - review & editing. **Yosie Andriani:** Conceptualization, Methodology, Writing - review & editing. **Mohd Effendy Abd Wahid:** Resources, Writing - review & editing. **Min Pau Tan:** Writing - review & editing. **Muhd Danish-Daniel:** Writing - review & editing. **Li Lian Wong:** Writing - review & editing. **Tengku Sifzizul Tengku-Muhammad:** Writing - review & editing. **Wen Jye Mok:** Writing - review & editing. **Patrick Sorgeloos:** Writing - review & editing. **Yeong Yik Sung:** Conceptualization, Methodology, Project administration, Writing - original draft, Writing - review & editing.

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